TASK 2 - Structural Load Analysis

Slab, beam, column + wind and earthquake loads | IS codes

1. Objective and structural system

This report presents the load calculation and analysis of the main structural elements of the G+1 building planned in Task 1: a two-way floor **slab S1** (living-room panel 4.5 x 5.0 m), an edge **beam B1** (span 5.0 m) and a ground-floor **column C1**, followed by basic-level **wind** and **earthquake** loads on the whole building. Design uses the limit-state method of IS 456:2000 with **M25** concrete (f_{ck} = 25 N/mm²) and **Fe500** steel (f_y = 500 N/mm²). Loads follow IS 875 (Parts 1-3) and IS 1893 (Part 1):2016; detailing follows SP 16 and IS 13920:2016. All assumptions are stated so that the values can be changed in a software model (STAAD.Pro / ETABS).

2. Design loads

Load	Value	Basis
Self weight of RCC	25 kN/m³	IS 875 Part 1
Slab self weight (160 mm)	4.0 kN/m²	0.16 x 25
Floor finish + screed + ceiling plaster (dead)	1.0 kN/m²	Assumed
Brick masonry (230 mm, plastered)	20 kN/m³ = 11.5 kN/m per 2.5 m wall	IS 875 Part 1
Live load - residential rooms	2.0 kN/m²	IS 875 Part 2
Live load - accessible roof	1.5 kN/m²	IS 875 Part 2
Design combinations	1.5(DL+LL); 1.2(DL+LL±EL); 1.5(DL±EL); 0.9DL±1.5EL	IS 456 Table 18

Total service load on floor slab w = 4.0 + 1.0 + 2.0 = **7.0 kN/m²**; factored load w_u = 1.5 x 7.0 = **10.5 kN/m²**.

3. Slab S1 - two-way slab (two adjacent edges discontinuous)

Effective spans l_x = 4.5 m, l_y = 5.0 m, so l_y/l_x = 1.11 < 2 and the slab acts in two-way action. Bending-moment coefficients are taken from IS 456 Table 26 (case 4) and interpolated: α_x(-ve) = 0.0538, α_x(+ve) = 0.0406, α_y(-ve) = 0.047, α_y(+ve) = 0.035. Overall depth D = 160 mm, effective depth d = 135 mm (span/depth = 33.3 < 26 x 1.4, OK).

Section	Mu (kN·m/m)	Ast req. (mm²/m)	Reinforcement	Ast prov. (mm²/m)
Short span, mid (+ve)	8.62	192 (min)	10φ @ 300 c/c	262
Short span, support (-ve)	11.43	201	10φ @ 300 c/c	262
Long span, mid (+ve)	7.44	192 (min)	8φ @ 260 c/c	193
Long span, support (-ve)	9.99	192 (min)	8φ @ 260 c/c	193

Moment capacity check: $M_{u,lim} = 0.138\, f_{ck}\, b\, d^2$ = 62.9 kN·m >> 11.4 kN·m (safe, under-reinforced). Minimum steel 0.12% bD = 192 mm²/m governs. Shear: $V_u = w_u l_x/2$ = 23.6 kN/m, τ_v = 0.17 N/mm² < τ_c = 0.32 N/mm² - no shear reinforcement needed. Torsion steel (0.75 A_{st} over l/5 at the two discontinuous corners) and 2 x 10φ extra top bars at edges are to be added as per IS 456 cl. D-1.8.

4. Beam B1 - 230 x 500 mm, simply-supported span 5.0 m (edge beam)

Loads on beam (factored): slab load transferred as equivalent UDL = $w_u l_x/6 \times [3 - (l_x/l_y)^2]$ = **17.2 kN/m**; self-weight = 0.23 x 0.5 x 25 x 1.5 = **4.31 kN/m**; 230 mm brick wall (2.5 m high) = 0.23 x 2.5 x 20 x 1.5 = **17.3 kN/m**. Total w_u = **38.8 kN/m**. Hence $M_u = w_u L^2/8$ = **121.3 kN·m** and $V_u = w_u L/2$ = **97.0 kN** (conservatively taken as simply supported; continuity will reduce the sagging moment).

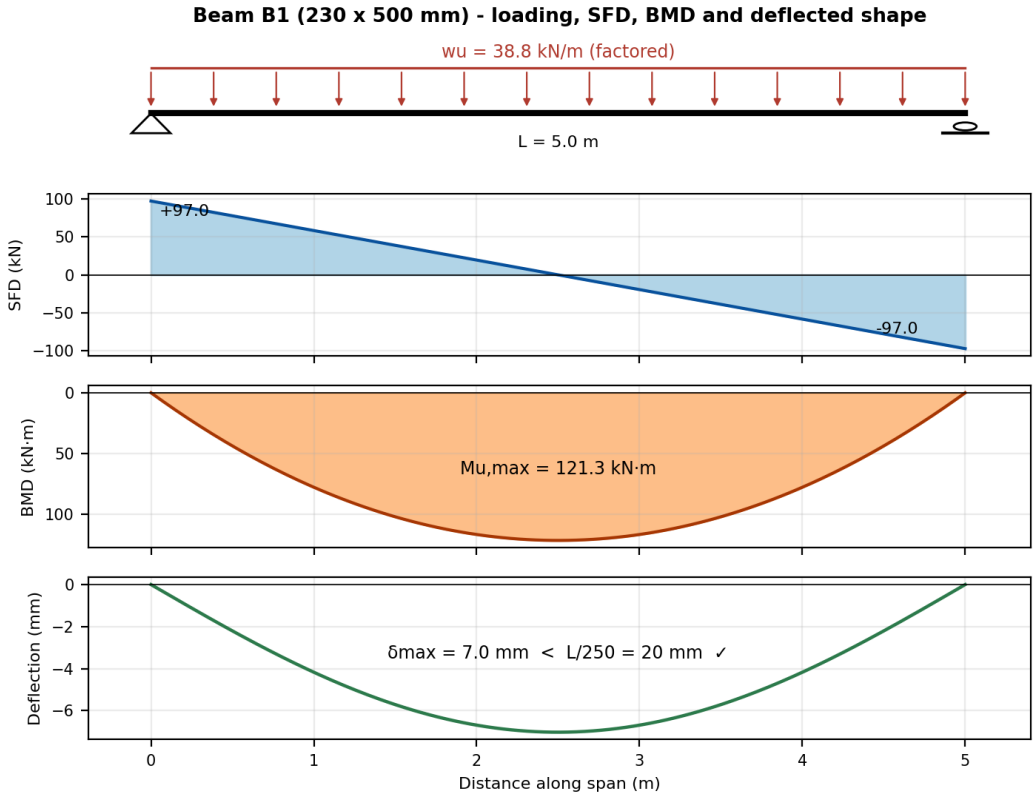


Fig. 1 Loading, shear-force diagram, bending-moment diagram and deflected shape of beam B1

Flexure: $M_{u,lim} = 0.138\, f_{ck}\, b\, d^2 = 167.9$ kN·m > $M_u = 121.3$ kN·m, so a singly reinforced section is adequate. A_{st} required = **698 mm²** → provide **4-16φ** ($A_{st} = 804$ mm², $p_t = 0.76\%$), min. steel $0.85bd/f_y = 180$ mm² OK. **Shear:** $\tau_v = 0.92$ N/mm² > $\tau_c = 0.57$ N/mm² (< $\tau_{c,max} = 3.1$), so stirrups are needed: $V_{us} = 36.4$ kN; spacing from strength = 552 mm, but maximum allowed = 300 mm → provide **2-legged 8φ @ 300 mm c/c** (closer, 8φ @ 100-150 mm, within 2d of supports for ductile detailing). **Serviceability:** span/depth = 10.9 < 20 x 1.3 = 26 OK; elastic deflection with cracked stiffness ($I_{eff} \approx 0.5\, I_g$) = 7.0 mm < L/250 = 20 mm OK.

5. Column C1 - 300 x 300 mm (corner/edge column, G+1)

Tributary area = 4.5 x 5.0 = 22.5 m². Axial service load from roof, first floor, beams, brick walls, parapet and self weight is **P = 645 kN**; factored $P_u = 1.5P$ = **968 kN** (detailed in the calculation sheet). Unsupported length 2.5 m, effective length 0.65 x 3.0 = 1.95 m, $l_{ex}/D = 6.5 < 12$ → short column; minimum eccentricity e_{min} = 15 mm ≤ 0.05D = 15 mm, so IS 456 cl. 39.3 applies: $P_u = 0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}$. With **4-16φ** ($A_{sc} = 804$ mm² = 0.89% > 0.8%), capacity = **1161 kN** > 968 kN (utilisation 83%). Lateral ties 8φ @ 250 mm c/c (least of 300, 16 x 16, 300 mm).

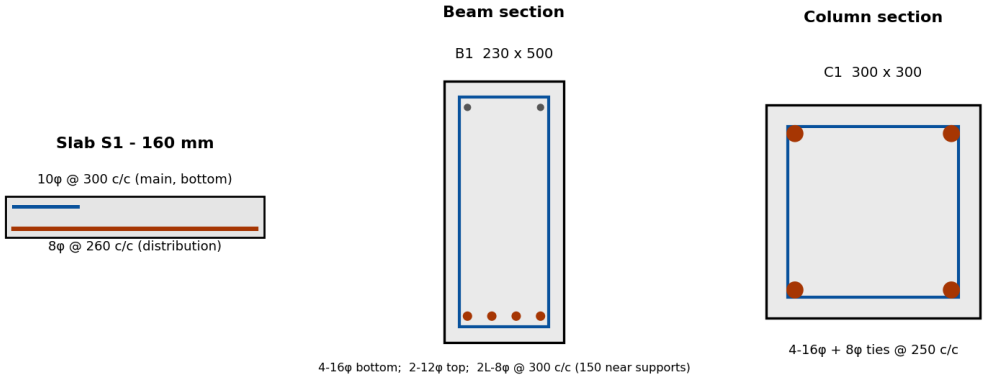


Fig. 2 Reinforcement details: slab strip, beam B1 and column C1

6. Lateral loads - wind and earthquake (basic level)

Wind (IS 875 Part 3): basic wind speed $V_b = 39$ m/s (assumed), $k_1 = k_2 = k_3 = k_4 = 1.0$ → $V_z = 39$ m/s; design pressure $p_z = 0.6V_z^2 = 0.913$ kN/m². With net pressure coefficient $C_p = 1.0$ (windward +0.7, leeward -0.3) on the wind-facing wall 10.73 m x 7.45 m = 79.9 m²: **F_w = 73.0 kN**, overturning moment = 272 kN·m.

Earthquake (IS 1893 Part 1:2016): Zone IV (Z = 0.24), I = 1.0, R = 5 (special moment frame), medium soil. Approximate period $T_a = 0.075h^{0.75} = 0.29$ s → $S_a/g = 2.5$. $A_h = (Z/2)(I/R)(S_a/g) =$ **0.060**. Seismic weight (DL + 25% LL): $W_1 = 1478$ kN, $W_2 = 1277$ kN, W = 2755 kN. Base shear $V_B = A_h W =$ **165.3 kN**, distributed as $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$: **$Q_1$ = 37.1 kN** (first floor), **Q_2 = 128.2 kN** (roof).

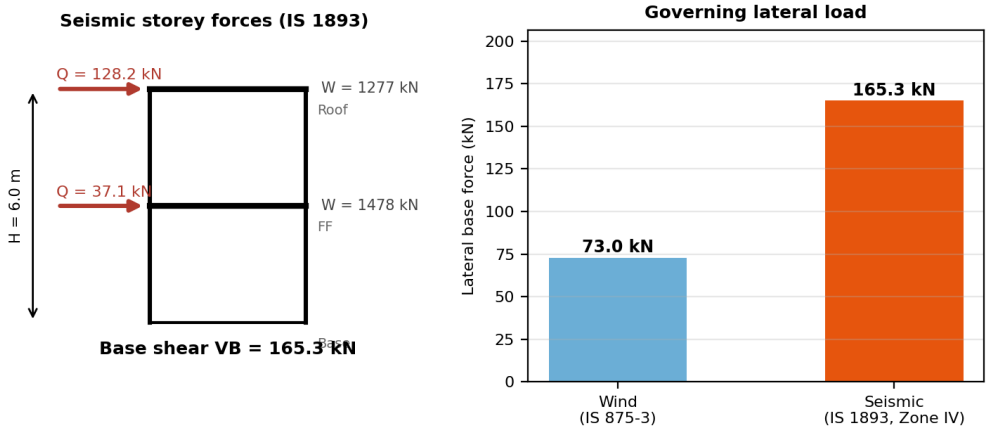


Fig. 3 Storey seismic forces and comparison of wind and seismic base force

7. Structural behaviour

Gravity load path: the slab carries floor loads in bending in both directions and passes them to the supporting beams as triangular (short side) and trapezoidal (long side) loads; beams carry these in bending and shear to the columns, which work mainly in axial compression and deliver the load to the footings and soil. The beam has a maximum sagging moment at mid-span (121 kN·m) and maximum shear at the supports (97 kN), which is why tension steel is concentrated at the bottom of the mid-span and stirrups are closer near the supports.

Lateral behaviour: the seismic base shear (165 kN) is about 2.3 times the wind force (73 kN), so **earthquake governs** the lateral design of this building. The rigid floor slabs act as diaphragms distributing the inertia forces to the frames, and the frames resist them through bending of beams and columns (sway). Approximately 78% of the base shear acts at the roof level because of the higher mass-height product, so the upper storey columns and the beam-column joints need ductile detailing (IS 13920). Storey drift must be limited to 0.004h = 12 mm (IS 1893 cl. 7.11.1.1) - to be verified in the STAAD/ETABS frame model.

8. Conclusion

The adopted sizes - slab 160 mm, beam 230 x 500 mm with 4-16φ bottom bars, column 300 x 300 mm with 4-16φ - satisfy the strength and serviceability checks under gravity loads, and the structure is governed by earthquake rather than wind for lateral loading. The next step for a real project would be a 3D frame analysis with seismic load combinations and verification of the strong-column/weak-beam requirement (see Task 4).

CALCULATION SHEET - Task 2

Step-by-step design calculations
(limit-state method, IS 456:2000)

A. Slab S1 (4.5 m x 5.0 m, D = 160 mm, d = 135 mm, cover 20 mm)

Step	Calculation	Result	Ref.
1	Dead load: slab 0.16 x 25 + finish 1.0	5.0 kN/m²	IS 875-1
2	Live load (residential)	2.0 kN/m²	IS 875-2
3	w = DL + LL = 5.0 + 2.0	7.0 kN/m²	
4	w_u = 1.5 x 7.0	10.5 kN/m²	IS 456 T.18
5	l_y/l_x = 5.0/4.5	1.111 (two-way)	cl. 24.4
6	$M_x(-)$ = 0.0538 x 10.5 x 4.5²	11.43 kN·m/m	Table 26
7	$M_x(+)$ = 0.0406 x 10.5 x 4.5²	8.62 kN·m/m	Table 26
8	$M_y(-)$ = 0.047 x 10.5 x 4.5²	9.99 kN·m/m	Table 26
9	$M_y(+)$ = 0.035 x 10.5 x 4.5²	7.44 kN·m/m	Table 26
10	$M_{u,lim}$ = 0.138 x 25 x 1000 x 135² / 10^6	62.9 kN·m (OK)	Annex G
11	A_{st}: 0.87 f_y A_{st} d [1 - $A_{st}f_y/(b\,d\,f_{ck})$] = M_u	see slab table, Task 2 §3	Annex G
12	$A_{st,min}$ = 0.12% x 1000 x 160	192 mm²/m	cl. 26.5.2.1
13	Spacing s = 1000 x a_φ / A_{st}; max = min(3d, 300)	10φ @ 300; 8φ @ 260	cl. 26.3.3
14	V_u = 10.5 x 4.5 / 2; $\tau_v = V_u/(b\,d)$	23.6 kN; 0.17 N/mm²	cl. 40.1
15	τ_c (p_t = 0.19%, M25)	0.32 N/mm² > τ_v (safe)	Table 19

B. Beam B1 (b = 230 mm, D = 500 mm, span L = 5.0 m)

Step	Calculation	Result	Ref.
1	Slab load (trapezoidal → UDL): $w_u l_x/6 \times (3 - (l_x/l_y)^2)$ = 10.5 x 4.5/6 x (3 - 0.81)	17.25 kN/m	IS 456 / std.
2	Self weight: 1.5 x 0.23 x 0.50 x 25	4.31 kN/m	
3	Wall: 1.5 x 0.23 x 2.5 x 20	17.25 kN/m	IS 875-1
4	w_u = 1 + 2 + 3	38.81 kN/m	
5	$M_u = w_u L^2/8$ = 38.81 x 5² / 8	121.3 kN·m	
6	$V_u = w_u L/2$	97.0 kN	
7	d = 500 - 25 - 8 - 8 (cover, stirrup, half bar)	≈ 460 mm	
8	$M_{u,lim}$ = 0.138 x 25 x 230 x 460² / 10^6	167.9 kN·m > M_u	Annex G
9	A_{st} from quadratic of M_u = 0.87 f_y A_{st} d (1 - $A_{st}f_y/bdf_{ck}$)	698 mm²	Annex G
10	Provide 4-16φ: 4 x 201.1	804 mm² (p_t = 0.76%)	
11	$A_{st,min} = 0.85\,b\,d / f_y$	180 mm² (OK)	cl. 26.5.1.1
12	$\tau_v = V_u/(b\,d)$	0.92 N/mm²	cl. 40.1
13	τ_c (p_t = 0.76%)	0.57 N/mm² ($\tau_{c,max}$ = 3.1)	Table 19/20
14	$V_{us} = V_u - \tau_c bd$	36.4 kN	cl. 40.4
15	$s_v = 0.87\,f_y\,A_{sv}\,d / V_{us}$ (A_{sv} = 100.5 mm²)	552 mm	cl. 40.4
16	$s_{v,max}$ = min(0.75d, 300, $0.87 f_y A_{sv}/0.4b$)	300 mm → provide 300	cl. 26.5.1.5/6
17	Deflection: $\delta = 5\,w_s\,L^4 / (384\,E_c\,0.5\,I_g)$; $E_c = 5000\sqrt{f_{ck}}$ = 25000 MPa, $w_s = w_u/1.5$ = 25.9 kN/m	7.0 mm < 20 mm	cl. 23.2

C. Column C1 (300 x 300 mm, M25 / Fe500)

No.	Axial load component	P (kN)
1	Roof slab DL (5.5 kN/m² x 22.5 m²)	123.8
2	Roof slab LL (1.5 kN/m² x 22.5 m²)	33.8
3	First-floor slab DL (5.0 kN/m² x 22.5 m²)	112.5
4	First-floor slab LL (2.0 kN/m² x 22.5 m²)	45.0
5	Beams, 2 floors (2.875 kN/m x 9.5 m x 2)	54.6
6	Brick wall, FF level + GF level (11.5 kN/m x 9.5 m x 2)	218.5
7	Parapet at roof (4.6 kN/m x 9.5 m)	43.7
8	Column self weight (2 storeys)	13.5
	Total service load P	**645.3**
	Factored P_u = 1.5 P	**968.0**

Step	Calculation	Result	Ref.
1	Slenderness l_{ex}/D = 0.65 x 3000 / 300	6.5 < 12 → short	cl. 25.1.2
2	e_{min} = l/500 + D/30 = 2500/500 + 300/30	15 mm ≤ 0.05D = 15 mm	cl. 25.4
3	$A_{sc,min}$ = 0.8% x 90000	720 mm²	cl. 26.5.3.1
4	Provide 4-16φ	804 mm² (0.89%)	
5	$P_{u,cap} = 0.4 f_{ck}(A_g - A_{sc}) + 0.67 f_y A_{sc}$	1161 kN > 968 kN	cl. 39.3
6	Lateral ties: min(300, 16 x 16 = 256, 300) → 8φ @ 250	OK	cl. 26.5.3.2

D. Wind load

Step	Calculation	Result	Ref.
1	V_b (basic wind speed, assumed)	39 m/s	IS 875-3 Fig. 1
2	$V_z = V_b k_1 k_2 k_3 k_4$ (all = 1.0)	39 m/s	cl. 6.3
3	$p_z = 0.6 V_z^2$	913 N/m² = 0.913 kN/m²	cl. 7.2
4	Exposed area A = 10.73 x 7.45	79.9 m²	
5	$F_w = C_p p_z A$ (C_p = 1.0)	73.0 kN	cl. 7.3
6	$M_{ot} = F_w$ x H/2	272 kN·m	

E. Earthquake load (equivalent static method)

Step	Calculation	Result	Ref.
1	Z = 0.24 (Zone IV), I = 1.0, R = 5.0 (SMRF), medium soil		IS 1893 T.3,8,9
2	$T_a = 0.075 h^{0.75} = 0.075 \times 6.0^{0.75}$	0.288 s	cl. 7.6.2
3	S_a/g (0.10 s ≤ T ≤ 0.55 s, medium soil)	2.50	Fig. 2
4	$A_h = (Z/2)(I/R)(S_a/g)$ = 0.12 x 0.2 x 2.5	0.060	cl. 6.4.2
5	W_1: slab 5.0x94.5 + 25%LL 0.25x2.0x94.5 + beams 198 + cols 81 + walls 678	1478 kN	cl. 7.4
6	W_2: slab 5.5x94.5 + beams + ½cols + ½walls + parapet 179	1277 kN	cl. 7.4
7	$W = W_1 + W_2$	2755 kN	
8	$V_B = A_h W$	165.3 kN	cl. 7.6.1
9	$Q_1 = V_B W_1 h_1^2 / \Sigma(W_j h_j^2)$ (h = 3 m)	37.1 kN	cl. 7.6.3
10	$Q_2 = V_B W_2 h_2^2 / \Sigma(W_j h_j^2)$ (h = 6 m)	128.2 kN	cl. 7.6.3
11	Overturning moment = Q_1x3 + Q_2x6	881 kN·m	

Note: the seismic zone (Z = 0.24) and basic wind speed (39 m/s) are assumed for illustration; for a real site the local code/zone map must be used. Values are basic-level hand calculations intended to be cross-checked with STAAD.Pro / ETABS.